

Spring REST Hands-on Exercises

Exercise 1: Create a Spring Web Project using Maven

The screenshot shows an IDE window with the 'Run' tab selected. The console output shows the following sequence of events:
1. `rtedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 26 with PID 17056 (C:\Users\User\Downloads\`
2. `rtedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"`
3. `rtedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to`
4. `rtedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to`
5. `rtedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)`
6. `rtedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]`
7. `rtedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]`
8. `rtedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 2179 ms`
9. `rtedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'`
10. `rtedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 4.197 seconds (process running for 5.438)`
The bottom status bar shows the file path `spring_learn > src > main > java > com > cognizant > spring_learn > SpringLearnApplication`, the time `7:14`, encoding `LF`, character set `UTF-8`, and 4 spaces. The system tray at the bottom shows a temperature of `31°C` and the date `14-07-2026`.

The screenshot shows an IDE window with the file `SpringLearnApplication.java` open. The console output is as follows:

```

:: Spring Boot ::      (v4.1.1-SNAPSHOT)

2026-07-14T15:14:01.185+05:30 INFO 17056 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLe
2026-07-14T15:14:01.193+05:30 INFO 17056 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile
2026-07-14T15:14:01.366+05:30 INFO 17056 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property
2026-07-14T15:14:01.366+05:30 INFO 17056 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional we
2026-07-14T15:14:03.418+05:30 INFO 17056 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialize
2026-07-14T15:14:03.445+05:30 INFO 17056 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service
2026-07-14T15:14:03.446+05:30 INFO 17056 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet
2026-07-14T15:14:03.546+05:30 INFO 17056 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicati
2026-07-14T15:14:04.360+05:30 INFO 17056 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started or
2026-07-14T15:14:04.381+05:30 INFO 17056 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLe

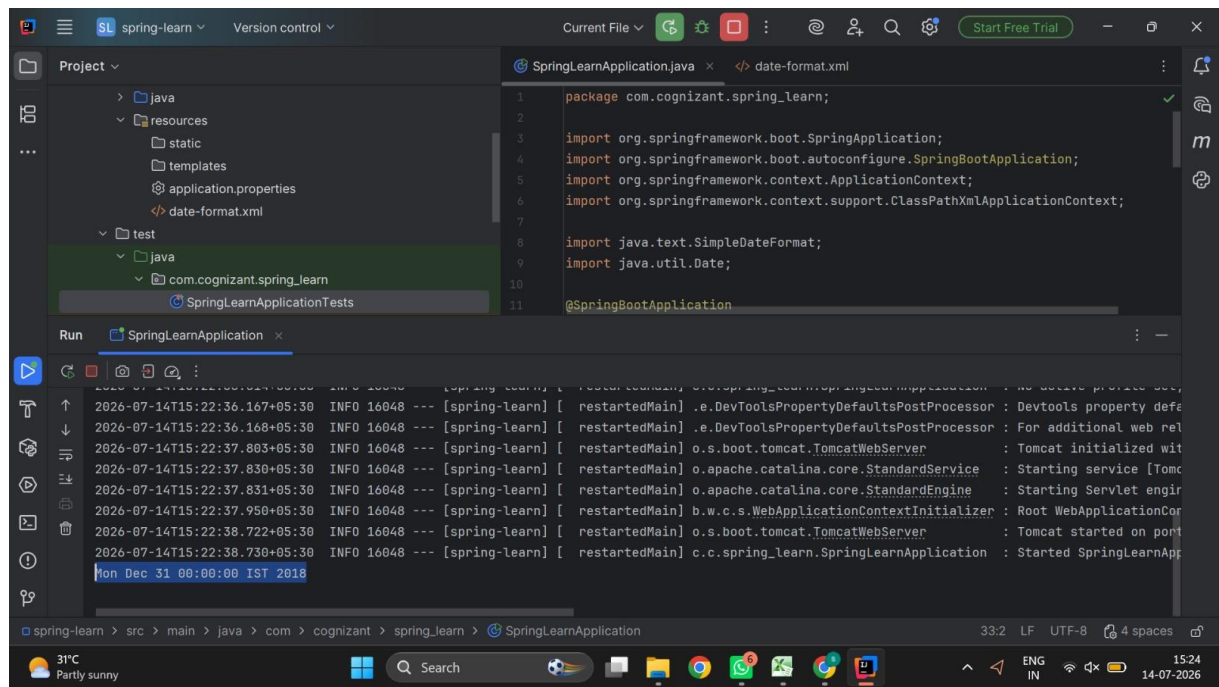
```

The IDE interface includes a project explorer on the left, a file explorer at the top, and a status bar at the bottom showing the current file path and system information.

Created a Spring Boot web project using Maven and successfully ran the application.

Verified that the embedded Tomcat server started successfully.

Exercise 2: Spring Core – Load Country from Spring Configuration XML



Configured a Spring XML file and loaded the bean using ApplicationContext.
Verified the output by successfully reading the configured data.

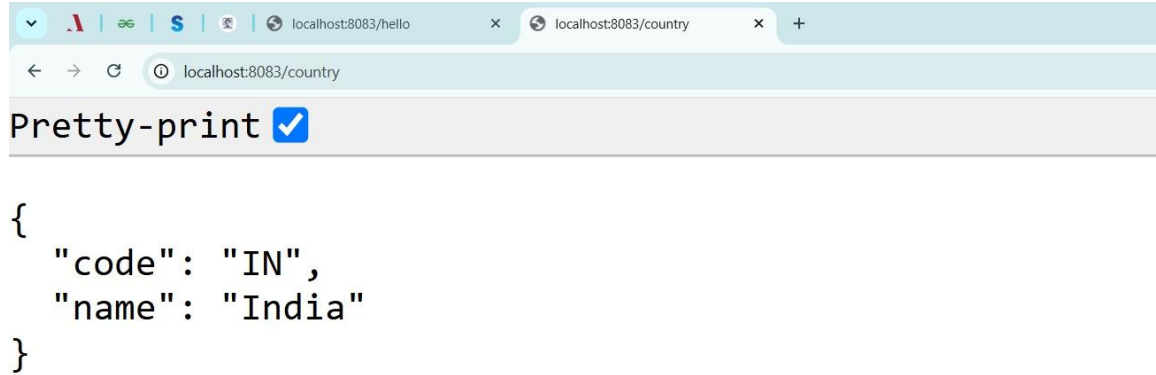
Exercise 3: Hello World RESTful Web Service

Implemented a simple RESTful web service using Spring Boot.
Verified the endpoint by displaying 'Hello World!!' in the browser



Hello World!!

EXERCISE 4 REST - Country Web Service



EXERCISE 5 REST - Get country based on country code

This REST service returns country details based on the country code provided in the URL using `@PathVariable`.

It searches the list of countries in `country.xml` and performs a case-insensitive match to return the requested country.

